MTH 101 – MATHEMATICS

LECTURE 1

Introduction to Sets

Study notes based on the topics and class discussion you described.

Lecture coverage

Course outline introduction and an introductory discussion of Set Theory, focusing on the meaning and notation of sets and the Roster and Set-Builder methods.

Note: Topics beyond the introductory set discussion are not presented as material already taught in this lecture.

1. Course Outline

Overview:

The course is scheduled to cover the following topics during the semester:

Topics:

- Set Theory
- Theory of Quadratic Equations
- Sequences and Series
- Mathematical Induction
- Permutation and Combination
- Binomial Theorem
- Trigonometry
- Complex Numbers

Scope of this lecture:

The class introduced the course outline and began Set Theory. The discussion focused on what sets are, how sets are represented, and the two methods introduced: the Roster Method and the Set-Builder Method. The other course-outline topics are listed for reference; they are not treated as topics taught in this lecture.

2. Introduction to Set Theory

Meaning of a set:

A set is a well-defined collection of distinct objects, called elements or members. The objects may be numbers, letters, names or other clearly specified items.

Meaning of well-defined:

A collection is well-defined when it is clear whether a particular object belongs to it or does not belong to it. For example, “the vowels in the English alphabet” is well-defined because membership can be checked objectively.

Examples of sets:

- The set of vowels in the English alphabet.
- The set of even natural numbers less than 10.
- The set of days of the weekend.

Examples of collections that are not well-defined:

- A collection of beautiful songs.
- A collection of very intelligent students.

Why these are not well-defined:

Words such as “beautiful” and “very intelligent” can mean different things to different people unless a clear rule is provided.

3. Basic Set Terminology and Notation

Set:

A collection usually represented by a capital letter such as A, B, C or X.

Element / member:

An object that belongs to a set.

Curly brackets:

Sets are commonly written inside braces { }. For example, A = {1, 2, 3}.

Belongs to:

The symbol $\in$ means “is an element of”. If A = {1, 2, 3}, then $2 \in A$.

Does not belong to:

The symbol $\notin$ means “is not an element of”. If A = {1, 2, 3}, then $5 \notin A$.

Equality of sets:

Two sets are equal when they contain exactly the same elements, regardless of the order in which the elements are written.

Order:

The order of elements does not affect a set. Thus {1, 2, 3} and {3, 2, 1} describe the same set.

Repetition:

Repeated listing does not create additional elements. Thus {1, 1, 2, 2, 3} represents the same set as {1, 2, 3}.

4. The Two Methods of Representing Sets

A. Roster Method (Listing Method):

The Roster Method represents a set by listing all its elements inside curly brackets. Each distinct element is written out.

Example 1:

Let A be the set of the first five natural numbers. In roster form: A = {1, 2, 3, 4, 5}.

Example 2:

Let B be the set of vowels in the English alphabet. In roster form: B = {a, e, i, o, u}.

Example 3:

Let C be the set of even natural numbers less than 10. In roster form: C = {2, 4, 6, 8}.

Features of the Roster Method:

- The elements are written out individually.
- It is convenient when a set has a small or limited number of elements.
- The elements are enclosed in curly brackets.
- Each distinct element is listed only once.

B. Set-Builder Method (Rule Method):

The Set-Builder Method describes a set by stating the property or rule that every element in the set must satisfy. Instead of listing every element, it gives a condition for membership.

General form:

A = {x : x satisfies a stated condition}. The vertical bar | or colon : is read as “such that”.

Example 1:

A = {x : x is a natural number and $1 \leq x \leq 5$}. This describes the set {1, 2, 3, 4, 5}.

Example 2:

B = {x : x is a vowel in the English alphabet}. This describes {a, e, i, o, u}.

Example 3:

C = {x : x is an even natural number and x < 10}. This describes {2, 4, 6, 8}.

Features of the Set-Builder Method:

- It states a defining property or condition.
- It is useful for large sets and sets with infinitely many elements.
- It uses a variable, commonly x, and a condition that determines membership.
- The condition must be clear enough to decide whether an object belongs to the set.

5. Comparing the Roster and Set-Builder Methods

Roster Method:

Lists the elements of the set directly.

Set-Builder Method:

Describes the elements by a rule or property.

Same set, two representations:

A = {2, 4, 6, 8} in roster form can be written as A = {x : x is an even natural number and x < 10} in set-builder form.

Choosing a method:

Use roster form when the elements can be listed conveniently. Use set-builder form when a rule describes the set more clearly, especially for large or infinite collections.

6. Worked Examples

Example 1: Write in roster form:

Let P be the set of natural numbers greater than 0 and less than 6. The numbers that satisfy the condition are 1, 2, 3, 4 and 5. Therefore, P = {1, 2, 3, 4, 5}.

Example 2: Write in set-builder form:

Let Q = {3, 6, 9, 12, 15}. These are the first five positive multiples of 3. One suitable representation is Q = {x : x = 3n, where n $\in$ {1, 2, 3, 4, 5}}.

Example 3: Decide membership:

If R = {a, e, i, o, u}, then e $\in$ R, while b $\notin$ R.

Example 4: Compare two sets:

If M = {1, 2, 3} and N = {3, 2, 1}, then M = N because they contain exactly the same elements.

7. Summary of Lecture 1

Main points:

- The course outline identifies eight major areas to be covered in MTH 101.
- The first class introduced Set Theory.
- A set is a well-defined collection of distinct objects.
- The objects in a set are called elements or members.
- Sets are commonly represented using capital letters and curly brackets.

- The symbol $\in$ means “belongs to”, while $\notin$ means “does not belong to”.
- The Roster Method lists the elements of a set.
- The Set-Builder Method describes a set using a defining condition or rule.

8. Revision Questions

Questions:

- What is a set?
- What does it mean for a collection to be well-defined?
- What are the elements of a set?
- Which brackets are commonly used to represent sets?
- Explain the meaning of the symbols $\in$ and $\notin$.
- State the two methods of representing sets discussed in the lecture.
- Define the Roster Method and give one example.
- Define the Set-Builder Method and give one example.
- Write the set of the first four natural numbers in roster form.
- Express the set {2, 4, 6, 8} in set-builder form.
- Are {a, b, c} and {c, b, a} equal sets? Explain briefly.

Suggested answers:

- A set is a well-defined collection of distinct objects.
- Membership can be decided clearly and objectively.
- Elements or members are the objects contained in a set.
- Curly brackets { }.
- $\in$ means belongs to; $\notin$ means does not belong to.
- Roster Method and Set-Builder Method.
- Roster Method lists the elements, e.g. {1, 2, 3}.
- Set-Builder Method states a rule, e.g. {x : x is a natural number and $1 \leq x \leq 3$}.
- {1, 2, 3, 4}.
- {x : x is an even natural number and $2 \leq x \leq 8$}.
- Yes. They contain exactly the same elements; order does not matter.